

Journal of New Music Research

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/nnmr20

FluidHarmony: Defining an equal-tempered and hierarchical harmonic lexicon in the Fourier space

Gilberto Bernardes, Nádia Carvalho & Samuel Pereira

To cite this article: Gilberto Bernardes, Nádia Carvalho & Samuel Pereira (2022) FluidHarmony: Defining an equal-tempered and hierarchical harmonic lexicon in the Fourier space, Journal of New Music Research, 51:2-3, 142-161, DOI: 10.1080/09298215.2023.2202641

To link to this article: https://doi.org/10.1080/09298215.2023.2202641

© 2023 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

View supplementary material

Published online: 16 Jun 2023.

Submit your article to this journal

Article views: 247

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=nnmr20

JOURNAL OF NEW MUSIC RESEARCH
2022, VOL. 51, NOS. 2–3, 142–161
https://doi.org/10.1080/09298215.2023.2202641

Routledge
Taylor & Francis Group

OPEN ACCESS Check for updates

FluidHarmony: Defining an equal-tempered and hierarchical harmonic lexicon in the Fourier space

Gilberto Bernardes [a,b], Nádia Carvalho [a,b] and Samuel Pereira [a,b]

[a]University of Porto–Faculty of Engineering, Porto, Portugal; [b]INESC TEC, Porto, Portugal

ABSTRACT
FluidHarmony is an algorithmic method for defining a hierarchical harmonic lexicon in equal temperaments. It utilizes an enharmonic weighted Fourier transform space to represent pitch class set (pcsets) relations. The method ranks pcsets based on user-defined constraints: the importance of interval classes (ICs) and a reference pcset. Evaluation of 5,184 Western musical pieces from the 16th to 20th centuries shows FluidHarmony captures 8% of the corpus's harmony in its top pcsets. This highlights the role of ICs and a reference pcset in regulating harmony in Western tonal music while enabling systematic approaches to define hierarchies and establish metrics beyond 12-TET.

ARTICLE HISTORY
Received 26 October 2021
Accepted 5 April 2023

KEYWORDS
Harmony; discrete Fourier transform; pitch class sets; interval classes; lexicon

1. Introduction

Western art music is typically taught as a set of principles emerging from the common-practice tonal music period. These principles have been systematizred in the large plethora of musical treatises (Huron, 2016; Susanni & Antokoletz, 2012), which define *what* musical objects should be adopted and *how* they exist and relate in time. Typically, the journey commences with the study of multiple constructs at different hierarchies, such as scales, intervals, and prototypical chords. Once these constructs are mastered, functional harmony is studied, which ultimately dictates pitch relations and harmonic motion. In what follows, we adopt a long-standing linguistic metaphor (Swain, 1997) to refer to multiple constructs and functional harmony as the lexicon[1] and syntax of a musical grammar, respectively.

Throughout the common practice period, roughly from mid-seventeenth to late-nineteenth centuries, the Western tonal music lexicon and syntax remained rooted in the same fundamental principles (Aldwell et al., 2018; Rohrmeier & Pearce, 2018). In the twentieth century, the fundamentals of the common practice have been criticised, disrupted, and expanded to a degree without historical precedent.[2] Representative examples are the emancipation of the dissonance (Parncutt & hair, 2011; Tenney, 1988) and the rise of a myriad of individual harmonic systems replacing the prevailing functional tonality (Persichetti, 1961). The rise of novelty as an aesthetic value impelled composers to design new harmonic systems at an unprecedented rate. Thereupon, new harmonic treatises have been written by music composers and theorists. Notable examples include Messiaen's (1944) *La technique de mon langage musical*, Schoenberg's twelve-tone technique (Simms, 2000), Forte's (1973) set theory, and Perle's (1977) twelve-tone tonality.

Equal-tempered divisions of the octave beyond the traditional 12-tone equal temperament (TET) have also been explored to a great extent. The 24-TET division has been of interest to twentieth-century composers

CONTACT Gilberto Bernardes gba@fe.up.pt University of Porto–Faculty of Engineering, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal; INESC TEC, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal

Supplemental data for this article can be accessed here. https://doi.org/10.1080/09298215.2023.2202641

[1] We will often use the term lexicon, a word pertaining to linguistics, as a metaphorical manifestation in music. In linguistics, lexicon is used to describe the entire set of words from a language to which meaning is attributed. A direct parallel in music can lead to the understanding of a harmonic lexicon as all possible combinations of pitch. However, within the scope of this article, as in related literature, the term is understood as a variable and relatively small group of pitch class sets that establish the primal harmonic material to be used in a given context (e.g., a composition or a movement).

[2] It is certain that many of these new composition systems, such as integral serialism, have sought a highly democratic use of different pitch classes and even musical elements such as rhythm or dynamics. However, other composers of the 20th and 21st centuries seek a hierarchical use of pitches, which reinforces the need to propose a system that assists the composer in defining these hierarchies.

© 2023 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

such as Julián Carrillo, Ben Johnston, Harry Partch, Horațiu Rădulescu, Karlheinz Stockhausen, James Tenney, Ivan Wyschnegradsky, and La Monte Young (Anders & Miranda, 2010). As these multiple tunings and sonorities find their way into composition, a vast uncharted musical territory is available to composers.

Computation tools play a vital role in navigating this unfamiliar landscape in two prominent directions. First, sound synthesis methods, notably in the digital domain, foster the aural exploration of any tuning system. Second, the computer-aided algorithmic composition (CAAC) method supports the definition of hierarchical harmonic constructs. Existing systems typically address both areas. Representative software includes Csound (Boulanger, 2000), SuperCollider (McCartney, 2002), Max (Puckette, 2002), Bach and Cage libraries for the Max environment (Ghisi & Agostini, 2017), Pure Data (Puckette, 1996), OpenMusic and PWGL (Assayag et al., 1999; Laurson et al., 2009), JMSL (Didkovsky & Burk, 2001; Polansky et al., 1990), Common Music (Taube, 1997), and Strasheela (Anders & Miranda, 2010).

Comprehensive music theories beyond the 12-TET are challenging to model with the above systems; rather, they focus primarily on scales' definition. Strasheela (Anders & Miranda, 2010) is a notable exception, which applies constraint programming to define higher-level pitch-related concepts such as chords and scales from user-defined relations between pitch, pitch classes, and chord or scale degrees. The formalisation of the approach tackles the problem of defining equal-temperament higher-level constructs systematically. However, the user-driven parameterisation as heuristic constraints is arguably complex, defying the need for an automatic computational approach.

In this paper, we propose FluidHarmony, a CAAC method for defining a harmonic lexicon of pitch class sets (pcsets) in any equal-tempered division of the octave. The contributions of the proposed method can support composers in (1) prescribing a harmonic lexicon of pcsets within a given composition context at multiple hierarchies (or cardinalities); (2) establishing a harmonic hierarchy according to two user-defined constraints regulating the interval content and the pcset region of the lexicon; (3) exploring a wide array of harmonic spaces in any number of p-tone subdivisions of the octave; (4) guiding the composition process through providing aural feedback as well as topological and mathematical hierarchical representations that emerge from the Fourier representational space.

FluidHarmony computes a lexicon of pcsets in an equal-tempered, enharmonic Fourier space resulting from the discrete Fourier transform (DFT) of pcsets (Amiot, 2016; Lewin, 1959, 2001; Quinn, 2006, 2007). Mathematically, this framework is similar to the application of the DFT on audio signals. However, the signal space should be interpreted as the pitch class circle, and the signal itself as the weights given to the pitch classes in a set (Yust, 2019). For a comprehensive pedagogy on the application and interpretation of the DFT to pitch class distribution, please refer to Amiot (2016), Noll (2019), and Yust (2015b). The latter space has been shown to capture and quantify musical theoretical principles with unforeseen accuracy in voice-leading (Tymoczko, 2008), tonal regions modeling and subset structure (Bernardes et al., 2016, 2017; Yust, 2015a, 2015b), and the study of tuning systems (Amiot, 2016; Callender, 2007). Two properties of the Fourier space relevant to our method are the possibility of capturing interval class[3] (IC) and common pitch class relations from metrics in the Fourier magnitude and phase, respectively. Both magnitude and phase can be independently defined, fostering a method where a pcset lexicon results from the intersection of a reference set of pitch classes (or a vector defining the importance of each pitch class) with a reference IC vector imposing the importance of intervals.

Underlying our method lies an original contribution that enables the algebraic definition of weights to regulate the importance of ICs in the Fourier magnitude space. Furthermore, a pcset in the weighted Fourier space, $T(k)$, exists in a limited space due to its L_2 normalisation by cardinality, allowing pcsets with different cardinalities (or multiple hierarchies, such as notes, chords, and scales) to be represented and compared (Bernardes et al., 2016).

After populating the weighted Fourier space with all unique pcsets, within $1 \leq t \leq p$ cardinality, where $p \in \mathbb{Z}^+$ stands as the number of equal-tempered divisions of the octave, a harmonic lexicon is defined as the combination of two spatial constraints to be maximised: (1) pcset magnitude and (2) phase similarity to a reference pcset R, as illustrated in Figure 1. Larger pcset magnitudes indicate greater compliance to the user-defined IC distribution. Phase (or cosine) similarity to a reference pcset indicates greater compliance to a user-defined region. From the resulting harmonic lexicon and its underlying (magnitude and phase) metrics, we propose three hierarchical representations of pitch structures inspired by the concept of *event hierarchy* by Bharucha (1984a, 1984b) – a cognitive pitch theory that aims at tackling some limitations on the prevailing tonal pitch models within cognitive psychology (Krumhansl, 1979; Lerdahl, 1988;

[3] The concept of IC, also known as unordered pitch-class interval, has been coined within musical set theory (Forte, 1973) and extensively applied in atonal music to denote the smallest (interval) distance in the pitch class space between two unordered pitch classes. For example, the interval class between pitch classes 0 and 7 is IC5 as $0 - 7 = -7 \equiv -7 (mod 12) = 5$ is smaller than $7 - 0 = 7$, yet it accounts for both the complementary intervals of perfect fourth and fifth.

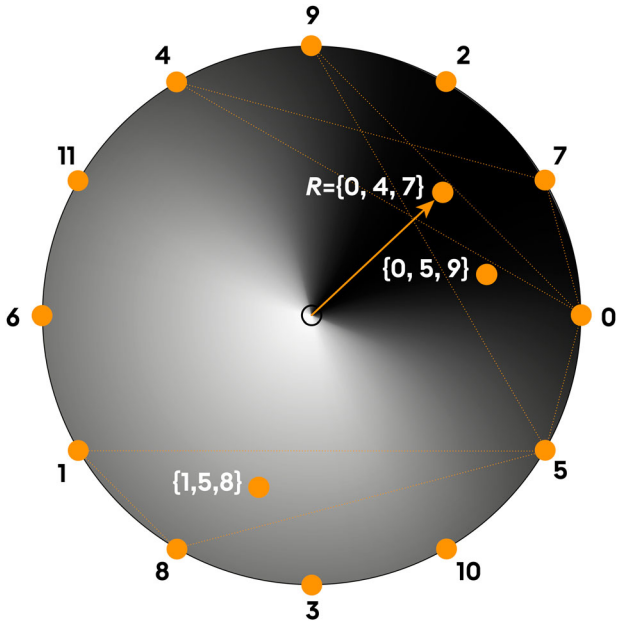


Figure 1. Caption: Illustration of a pcset lexicon definition in fifth Fourier coefficient of $\mathbb{Z}_{12}$, particularly relevant for tonal music. All pitch classes are represented in the circle along with three triads, one being the reference pcset $R = \{0, 4, 7\}$, highlighted by an orange directional vector. Stable pcsets occupy the darker shaded areas, which result from two main constraints: (1) larger magnitude values – i.e. the area close to the circumference, and (2) similar phase to a reference pcset *R*. Please note the proximity of the pcset {0, 5, 9} to *R*, as opposed to the pcset {1, 5, 8}–the transposition of *R* by a semitone, which in the context of tonal music is a very distant triad to *R*. Furthermore, in the neighbourhood of *R*, we can find its component pitch classes along with related diatonic (and consonant) pitch classes.

Longuet-Higgins, 1987; Shepard, 1982).[4] Our three hierarchies define (1) a pcset ranking denoting the harmonic stability of each pcset in the lexicon; (2) a topological representation of the Fourier (phase) space exposing pitch class retention between pcsets; and (3) a 'well-formed' and stratified space including pcsets of multiple cardinalities, following an earlier proposal by Lerdahl (2001).

We evaluate our research by conducting a systematic assessment of the FluidHarmony method's effectiveness in eliciting the lexica of a large corpus from Western art music. Our expectation is that both constraints are fundamental to the lexicon formation and the FluidHarmony ranked lexica mirror the Western music practice. By validating the proposed method in regulated practices, we aim to veridically assess how well the model captures existing musical practices and foster the exploration of less systematized harmonic systems, namely tuning systems beyond 12-TET.

The remainder of this paper is structured as follows. Section 2 reviews the musictheoretical value of the DFT of pcsets by unpacking information from the Fourier magnitude and phase. Section 3 introduces the mathematical definition of an ICweighted Fourier space, where a pcset's magnitude informs its compliance to a user-defined IC distribution. Section 4 details the definition of a harmonic lexicon as spatial constraints in a weighted Fourier space. Section 5 proposes three hierarchical pitch representations: a ranked pitch stability hierarchy, a topological common pitch class space, and a well-formed stratification space. Section 6 presents the evaluation of our FluidHarmony system in eliciting the harmonic lexica of a large corpus of Western art music. Finally, Section 7 presents the conclusions of our study, particularly unveiling the compositional implications of our FluidHarmony method, pointing towards the future endeavours of the research.

2. The discrete Fourier transform of equal-tempered pitch class sets

From the allusion to the DFT within the context of Lewin's interval function or IFUNC–cataloging the type and number of directed pitch class intervals between two pcsets (Lewin, 1959)–and the fundamental work of Quinn (2004) on the study of the DFT magnitude for music analysis, it came to be known that the Fourier space elicits many properties of music-theoretic value.

To map a pcset into the Fourier space, we first create a pcset distribution, $c(n)$, where $n = p$ $(p \in \mathbb{Z}^+)$, i.e. a distribution with n elements or the number of equal-tempered divisions of the octave. The distribution $c(n)$ is then normalised to unit sum (L_2 norm), $\bar{c}(n)$. The adoption of L_2 norm fosters a metrical space where multiple harmonic hierarchies or pcsets of any cardinality can be interpreted and compared. Finally, we apply Eq. 1 to compute its DFT, $T(k)$, such that:

$$T(k) = \sum_{n=0}^{N-1} \bar{c}(n)\, e^{\frac{-j2\pi kn}{N}}, \quad k \in \mathbb{Z}$$

$$\text{with} \quad \bar{c}(n) = \begin{cases} \frac{c(n)}{\|c(n)\|}, & \text{if } \|c(n)\| > 0 \\ c(n), & \text{otherwise} \end{cases} \qquad (1)$$

[4] A leading theory on tonal hierarchies is by Krumhansl (1979), which focus on cyclic properties of musical intervals within the octave. Her tonal hierarchy theory depicts the tonic as the most stable tone and the fifth and thirds consonant members of the major diatonic set as the next-most stable. The remaining tones in the diatonic set are less stable, and non-diatonic members are deemed least stable. Despite its remarkable explanation of tonal phenomena, Krumhansl's (1979) theory fails to explain essential phenomena from tonal music composition practice, such as temporal phenomena, and even less when applied to atonal or any equal temperament other than the 12-TET. Please refer to Butler (1990) for a detailed critique of the tonal hierarchy.

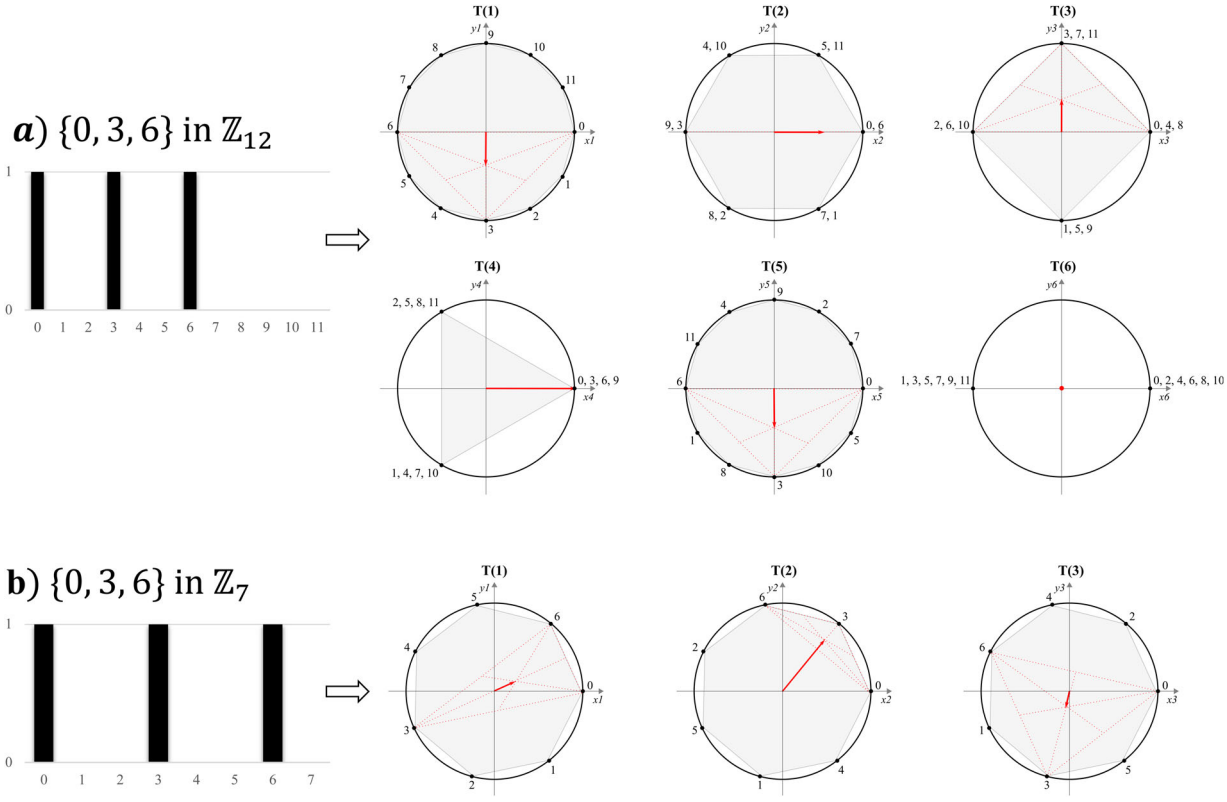


Figure 2. Caption: Representations of the {0, 3, 6} pcset as a distribution in the pitch class space (left images) and the Fourier space (right images). Examples (a) and (b) project the same pcset in 12- and 7-TET. Dashed (red) lines indicate the $T(k)$ property as a convex combination of pitch class components.

where $N = p$, i.e., the number of elements of the pcset distribution, and k is an integer number set to $1 \leq k \leq \frac{p}{2}$ for $T(k)$, representing each of the DFT coefficients. In the Fourier space, a $\frac{p}{2}$-element complex vector corresponding to the $1 \leq k \leq \frac{p}{2}$ DFT coefficients is adopted. The first DFT coefficient, $T(0)$, exposing the cardinality of a pcset, and all remaining symmetrical coefficients are excluded.[5]

Figure 2 shows the pcset distribution and the DFT of the {0, 3, 6} triad in both $\mathbb{Z}_{12}$ and $\mathbb{Z}_7$. To plot the multidimensional DFT of pcsets, we follow a visualisation strategy adopted in Harte et al. (2006) and Bernardes et al. (2016), which maps each Cartesian (real and imaginary) complex number per coefficient to the x- and y-axis. Furthermore, each coefficient location can also be defined in vector magnitude and angle by converting them to polar coordinates. Typically, in related music theory literature, the latter coordinate system is adopted (Amiot, 2013, 2016; Quinn, 2004; Yust, 2015a, 2015b). Furthermore, relations among pcsets, typically computed and quantified by distance or similarity metrics, are addressed in each of the magnitude or phase spaces, whose interpretation we detail next in Sections 2.1 and 2.2, respectively.

2.1. Fourier magnitude of pitch class sets

The magnitude of pcsets in the Fourier space, i.e. their size, has been used to study the shape of a pcset distribution as it quantifies how a given Fourier coefficient is $\frac{N}{k}$ periodic. Interpretations of Fourier magnitudes have promoted the study of pcsets' interval content based on their transposition-invariant representation (Amiot, 2016).

The pioneering work of Quinn (2004) interprets the Fourier magnitudes of pcsets on a coefficient basis. Quinn (2004) identified for each Fourier coefficient a prototypical pcset with maximal magnitude among remaining pcsets of the same cardinality, referred to as a *maximally even* (ME) set.[6] Furthermore, based on these

[5] When performing an N-point DFT on a pcset distribution, an N separate complex DFT output is returned. However, only the first $N/2 + 1$ coefficients are independent. The remaining coefficients $T(N/2 + 1)$ to $T(N - 1)$ are conjugate symmetric, thus provide no additional information about the spectrum of the pcset (Shannon, 1949).

[6] Please note that DFT has been used to generalize a method to find *perfectly balanced* sets in a microtonal universe that divides the octave into any k equal parts (Milne et al., 2015). The idea of *perfectly balanced* sets appears in complementarity with *perfectly even* sets (Clough & Douthett, 1991). Essentially, in a 12-tone equal temperament represented around the periodic circle, any set of equidistant pitch classes (e.g., the notes of a whole-tone scale) is a *perfectly even* set. On the other hand, any set of pitch classes whose mean position is the centre of the circle is said to be a *perfectly balanced* set (e.g., the notes of an octatonic scale are a *perfectly balanced* and *uneven* set). However, these two scales are *periodic* in that when they have

Table 1. Interpretation of Fourier coefficients' magnitude $T(1)$ to $T(6)$ for $\mathbb{Z}_{12}$. Maximal event (ME) sets and interval content (IC) per DFT coefficient are identified, i.e. prototype pcsets and complementary intervals with maximal magnitude. For enhanced readability, pcset prototypes are presented in their prime form (i.e. transposed to begin on 0); however, pcset prototypes can have up to p transpositions.

T(k)	ME set	IC	Harmonic quality
T (1)	0, 1	IC1	chromaticity
T (2)	0, 6	IC6	dyadicity or 'quartal quality'
T (3)	0, 4, 8	IC4	triadicity or hexatonicity
T (4)	0, 3, 6, 9	IC3	octatonicity
T (5)	0, 5	IC5	diatonicity
T (6)	0,2,4,6,8, 10	IC2	wholetone

pcset prototypes, Quinn (2004) establishes an association between individual DFT coefficients and ICs: $T(1) \leftrightarrow IC1$, $T(2) \leftrightarrow IC6$, $T(3) \leftrightarrow IC4$, $T(4) \leftrightarrow IC3$, $T(5) \leftrightarrow IC5$, and $T(6) \leftrightarrow IC2$. For example, the pcset $\{0, 3, 6, 9\}$ in $\mathbb{Z}_{12}$ is 3-periodic; therefore, its magnitude in $T(4)$ is maximal.[7] A pcset with comparatively large $T(4)$ can be associated with the IC3 (i.e. the complementary intervals of a minor third and a major sixth), such as $\{0, 1, 3, 6, 9\}$, which achieves the largest value in $T(4)$ among pcsets with cardinality $t = 5$.

Amiot (2017) has criticised the previous Fourier coefficient interpretation in showing that it better relates to the harmonic quality of a particular pcset, such as its level of diatonicity, octatonicity, and whole-toneness. For example, consider the sets $A = \{0, 4, 5\}$ and $B = \{0, 2, 4\}$. Conversely to pcset B, pcset A embeds an instance of IC5; therefore, we expected a larger $T(5)$. However, the magnitude of pcset B in $T(5)$, $\|T_B(5)\|^2 = 4$ is larger than $\|T_A(5)\|^2 = 2$, as $T(5)$ in many cases do not indicate the 'fifthyness' of a pcset so much as its diatonicity, and B is a more characteristic diatonic subset than A. For a greater discussion on the interpretation of these individual coefficients, please refer to Amiot (2017) and Callender (2007). Table 1 summarises the correspondence between Fourier coefficients $T(k)$ with pcset prototypes, ICs, and harmonic qualities in $\mathbb{Z}_{12}$.

Bernardes et al. (2016) build upon the IC interpretation of the DFT magnitude to propose a $\mathbb{Z}_{12}$ Fourier-weighted Tonal Interval Space. A weights vector regulates the importance of DFT coefficients to promote a space reflecting the hierarchy of ICs within Western tonal music. For example, the DFT coefficients $T(5)$ and $T(3)$ associated with IC5 and IC4 have larger weight values due to the fundamental importance of the intervals of perfect fourth/perfect fifth and major third/minor sixth in Western tonal harmony. This aspect is perfectly consistent with the results of Yust (2017), which point to the preponderance of $T(5)$, $T(3)$ and $T(2)$ in tonal music, where the compositional practice largely favours IC5, IC4 and IC6, respectively. We can observe, both in major and minor modes, a relation between DFT coefficients such as $T(5) < T(3) < T(2)$, and specifically what the author calls the 'tonal index': a DFT fingerprint to tonal pieces in which $T(2) + T(3) - T(5) \approx 0$ (Yust, 2017). In Bernardes et al. (2016), weights were computed using a brute-force approach to convey a ranking order of dyads and triads consonance from empirical ratings. In the Tonal Interval Space, the Fourier magnitude is interpreted as the combination of all coefficients and quantifies the pcset consonance as the degree to which a given pcset relates to the IC-distribution weights.

2.2. Fourier phase of pitch class sets

The phase of pcsets in the Fourier space, i.e., their direction, captured music theorists' attention as it models aspects of tonal music with unforeseen accuracy. Conversely to the transpositional-invariant Fourier magnitude, highlighting relations between the interval content of pcsets, Fourier phases are sensitive to pcset transposition and have been used to study voice-leading (Tymoczko, 2010), tonal regions and hierarchical tonal relations (Bernardes et al., 2016, 2017), and tuning systems (Amiot, 2016; Callender, 2007).

Yust (2015b, 2016) presents a geometric space adopting two-coefficient phases from the $\mathbb{Z}_{12}$ DFT of pcsets. The resulting space is a Cartesian plane representation of a torus (Amiot, 2016). The Fourier phase space $\langle T(5), T(3)\rangle$ has been adopted to describe music of the nineteenth and twentieth centuries (Amiot, 2013; Yust, 2015a, 2015b, 2016). Trajectories and distances in the resulting $\langle T(5), T(3)\rangle$ Fourier phase space reflect the usual Riemannian (dual) Tonnetze structure and unveil subset structure.

Bernardes et al. (2016) have shown that the Fourier phase space captures relations across multiple tonal pitch hierarchies as the angular distance between two given $T_1(k)$ and $T_2(k)$ vectors, such that:

$$T_1(k) \cdot T_2(k) = \| T_1(k) \| \| T_2(k) \| \cos\theta. \qquad (2)$$

Diatonic sets for each region of the 24 major and minor tonalities or the categorical (subdominant and dominant) harmonic functions per region exist as fuzzy clusters in

been rotated at a specific angle, they exhibit the same notes as the original position (Milne et al., 2015). It turns out that in a 12-TET, there are only two geometric patterns that are *perfectly balanced*, *uneven*, and *irreducibly periodic*—[C,Eb,E,Ab,A], and [C,Db,E,F,G,Ab,B]. However, in a microtonal universe, the possibility of finding sets with these properties is, in the very limit, infinite, which is perfectly in line with the microtonal facet of *FluidHarmony*.

[7] We can find the coefficient where a given periodic interval exists by dividing the number of p-tone subdivisions of the octave by the period. In the current pcset example, the maximal magnitude in $T(4)$ results can be computed as $12/3 = 4$.

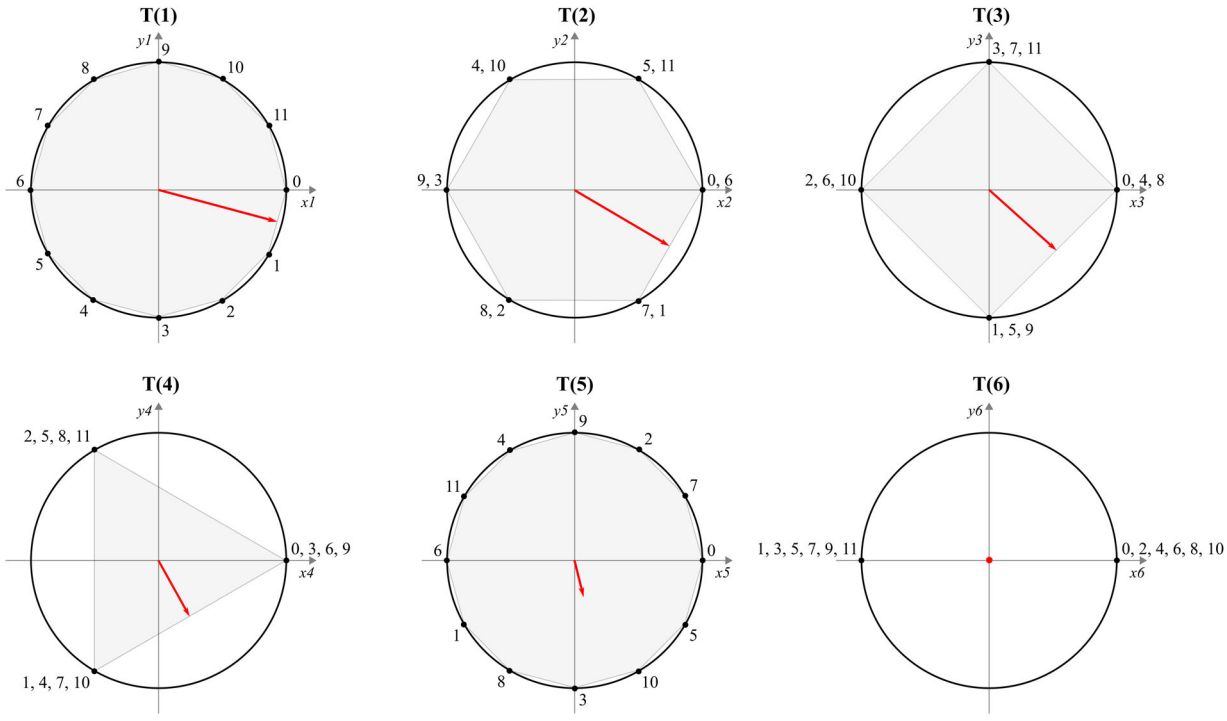


Figure 3. Visualisation of the magnitude of IC1 in the $\mathbb{Z}_{12}$ Fourier space. For each coefficient k, the real and imaginary parts of the DFT complex numbers are projected into the x- and y-axis, respectively.

the space. These distance relationships strive from common pitch class relations that are captured by smaller angular distances.

3. An interval content-weighted Fourier space

We propose an algebraic method to compute weights $w(k)$ for distorting the magnitude of the Fourier space, such that the magnitude of ICs in the Fourier space, $\|T_{IC}(k)\|$, equates with a user-defined IC distribution, I, with $\frac{p}{2}$ elements. For example, given an IC distribution $I = \{1, 2, 6, 8, 10, 4\}$ in $\mathbb{Z}_{12}$ (where each position defines the relative importance of an IC), $\|T_{IC1}(k)\|$ and $\|T_{IC5}(k)\|$ must be equal to 1 and 10, respectively. Conversely to the brute-force approach proposed in Bernardes et al. (2016) to find an optimal set of weights $w(k)$ that match empirical ratings of ICs' consonance, we propose an algebraic solution that drastically reduces the computational complexity involved in finding the (exact) weights $w(k)$ that match the user-defined IC distribution, I.

Translating the relative importance of ICs within the p-TET space into a set of Fourier weights $w(k)$ is a non-trivial problem due to the 'loose' links between Fourier coefficients and ICs (as previously discussed in Section 2.1). While the association between ICs and DFT coefficients, shown in Table 1, denotes maximal magnitude for pcsets embedding the respective IC, it does not hold in many cases due to the cyclic representation of the pitch classes in each coefficient. For example, a highly chromatic pcset cluster including all 12 pitch classes has zero magnitude in $T(1)$, the Fourier coefficient associated with IC1. Furthermore, all coefficients, irrespective of their association to a given IC, contribute to the magnitude of $\|T(k)\|$. Figure 3 illustrates the latter case by denoting the magnitude of the IC1 for coefficients k in a $\mathbb{Z}_{12}$ non-weighted Fourier space.

A square matrix, $M_{IC,k}$ with size $\frac{p}{2}$ can express the Fourier magnitude of all ICs per coefficient k. For example, Eq. 3 presents the matrix M for the $\mathbb{Z}_{12}$ Fourier space, where ICs and Fourier coefficients k are expressed as rows and columns, respectively.

$$\begin{pmatrix} \cos\frac{\pi}{12} & \cos\frac{\pi}{6} & \cos\frac{\pi}{4} & \cos\frac{2\pi}{6} & \cos\frac{5\pi}{12} & \cos\frac{\pi}{2} \\ \cos\frac{\pi}{6} & \cos\frac{2\pi}{6} & \cos\frac{\pi}{2} & \cos\frac{2\pi}{6} & \cos\frac{\pi}{6} & \cos(0) \\ \cos\frac{\pi}{4} & \cos\frac{\pi}{2} & \cos\frac{\pi}{4} & \cos(0) & \cos\frac{\pi}{4} & \cos\frac{\pi}{2} \\ \cos\frac{2\pi}{6} & \cos\frac{2\pi}{6} & \cos(0) & \cos\frac{2\pi}{6} & \cos\frac{2\pi}{6} & \cos(0) \\ \cos\frac{5\pi}{12} & \cos\frac{\pi}{6} & \cos\frac{\pi}{4} & \cos\frac{2\pi}{6} & \cos\frac{\pi}{12} & \cos\frac{\pi}{2} \\ \cos\frac{\pi}{2} & \cos(0) & \cos\frac{\pi}{2} & \cos(0) & \cos\frac{\pi}{2} & \cos(0) \end{pmatrix} \quad (3)$$

To regulate the importance of each IC in the baseline Fourier magnitude space, we compute the weights w algebraically as a linear system of matrix equations, such that:

$$w = M^{-1}I, \tag{4}$$

where I is a user-defined $\frac{p}{2}$-element distribution denoting the relative importance of ICs. Eq. 5 provides an example for computing the weights $w(k)$ from the IC distribution $I = \{1, 2, 6, 8, 10, 4\}$ in $\mathbb{Z}_{12}$.

$$\begin{pmatrix} \cos\frac{\pi}{12} & \cos\frac{\pi}{6} & \cos\frac{\pi}{4} & \cos\frac{2\pi}{6} & \cos\frac{5\pi}{12} & \cos\frac{\pi}{2} \\ \cos\frac{\pi}{6} & \cos\frac{2\pi}{6} & \cos\frac{\pi}{2} & \cos\frac{2\pi}{6} & \cos\frac{\pi}{6} & \cos(0) \\ \cos\frac{\pi}{4} & \cos\frac{\pi}{2} & \cos\frac{\pi}{4} & \cos(0) & \cos\frac{\pi}{4} & \cos\frac{\pi}{2} \\ \cos\frac{2\pi}{6} & \cos\frac{2\pi}{6} & \cos(0) & \cos\frac{2\pi}{6} & \cos\frac{2\pi}{6} & \cos(0) \\ \cos\frac{5\pi}{12} & \cos\frac{\pi}{6} & \cos\frac{\pi}{4} & \cos\frac{2\pi}{6} & \cos\frac{\pi}{12} & \cos\frac{\pi}{2} \\ \cos\frac{\pi}{2} & \cos(0) & \cos\frac{\pi}{2} & \cos(0) & \cos\frac{\pi}{2} & \cos(0) \end{pmatrix}^{-1} \times \begin{pmatrix} .1 \\ .2 \\ .6 \\ .8 \\ 1 \\ .4 \end{pmatrix} = \begin{pmatrix} -.665 \\ .069 \\ .579 \\ .231 \\ .608 \\ .099 \end{pmatrix} \tag{5}$$

The resulting weights $w(k)$ define the amount of distortion applied in each coefficient k to convey the degree of importance of the user-defined IC distribution I. The magnitude of a pcset in the weighted Fourier space V is computed as the weighted sum of the magnitude vector values, such that:

$$V = \sum_{k=1}^{M} w(k)\|T(k)\|, \tag{6}$$

where $M = \frac{p}{2}$ is the number of non-symmetrical Fourier coefficients for $\mathbb{Z}_p$ and $w(k)$ are weights that adjust the space to convey a user-driven IC distribution. The magnitude V of pcsets indicates the level of compliance to the user-defined IC distribution. Larger magnitude V values indicate greater compliance to the IC distribution I. In other words, larger magnitude values will be subset-class vectors of the IC distribution – i.e. sets embedded in the IC distribution, namely those IC with greater relative importance in the distribution.

4. Fluidharmony: A method for defining a harmonic lexicon in the Fourier space as spatial constraints

FluidHarmony is a CAAC method that adopts the proposed weighted-Fourier space to define a lexicon of pcsets in p-TET spaces. The resulting lexicon is regulated by two constraints in the Fourier space to be maximised: (1) the magnitude V of the $T(k)$ vectors to compute the quality of pcsets in complying with a user-defined IC distribution I and (2) the phase distance to a reference pcset $T_R(k)$ as the cosine similarity with any given $T_i(k)$ using Eq. 7 to define how well the pcset embeds a reference region.

$$C = \frac{T_1(k) \cdot T_2(k)}{\|T_1(k)\| \, \|T_2(k)\|} \tag{7}$$

Figure 4 shows the architecture of the method, namely its processing modules (rectangular blocks), user input parameters, and method output. Directional arrows indicate the flux of data across the multiple components. The input of the method includes four user-defined parameters: (1) a p-element IC distribution I, defining the relative importance of ICs; (2) a reference pcset R defining a region the lexicon occupies within the p possible transpositions;[8] (3) a list with all cardinalities $1 \leq t \leq p$ to be considered in the pcsets lexicon; (4) the balance α between the Fourier magnitude V and phase C constraints as a floating-point value in the $0 \leq \alpha \leq 1$ range, where zero and one indicate the sole adoption of magnitude and phase, respectively, and all real values within the interval indicate different degrees of importance between the constraints. The strength of defining a harmonic lexicon as a constraint-generation problem in the Fourier space lies in the flexible and integrated strategy to compute both the compliance to an IC and pcset spaces using the magnitude V and phase C information, respectively, and the capability to balance their importance via the α parameter.

Once the user-defined parameters are set, all pcsets P_i combinations within the $1 \leq t \leq p$ user-defined cardinalities populate the Fourier space using Eq. 1. For each pcset $T_i(k)$ in the Fourier space, we compute its weighted magnitude V_i (Eq. 6) and the phase to the reference pcset $T_R(k)$ using the cosine similarity C_i (Eq. 2). The weighted combination of the two metrics above defines the compliance to the constraints as a harmonic stability indicator H_i, such that:

$$H_i = \alpha \cdot V_i + (1 - \alpha) \cdot C_i \tag{8}$$

[8] The $\{0, 2, 4, 5, 7, 9, 11\} \in \mathbb{Z}_{12}$ is the pcset that defines the region of C major, whose 12 possible transpositions correspond to the regional space of the 12 major keys within tonal music. A reference pcset does not necessarily imply a tonal centre, but rather the pcsets which are privileged.

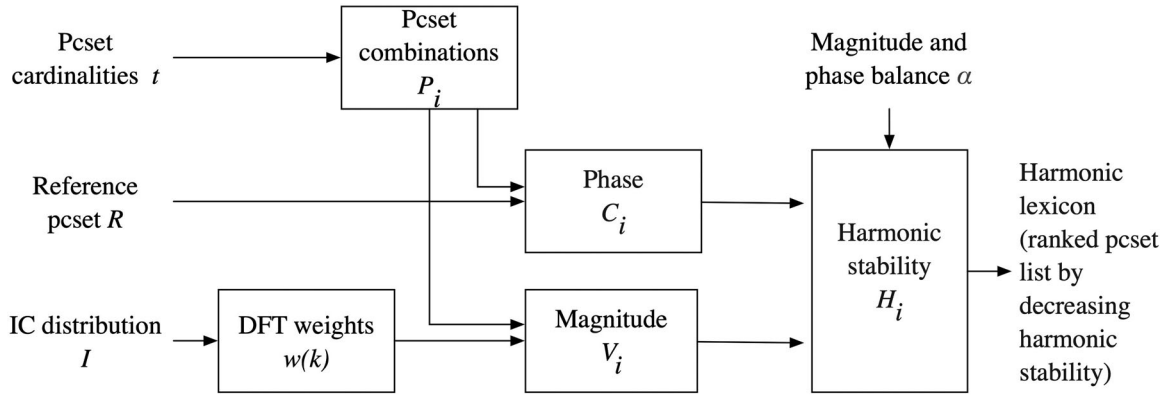


Figure 4. The FluidHarmony method's architecture includes input, processing modules (as rectangular blocks), and output. The arrow's direction indicates the data flux across the multiple components of the method.

Finally, the FluidHarmony method outputs a ranked list of pcsets P_i by decreasing H_i values. The larger the value, the greater the compliance to the user-defined harmonic lexicon constraints. The user makes the subjective decision of the number of pcsets in the lexicon to be adopted.

5. Eliciting harmonic event hierarchies and pitch-proximity relations

Based on the FluidHarmony metrics underlying our harmonic lexicon method, we propose three representations for defining event hierarchies[9] (an implicit pitch space): (1) a pcset ranking denoting the harmonic event stability of each pcset in the lexicon; (2) a topological representation in Fourier (phase) space exposing pitch class retention between pcsets; and (3) a stratification space where pcsets of multiple cardinality levels expose a 'well-formed' structure as proposed by Deutsch and Feroe (1981) and Lerdahl (2001).

The stability of a harmonic event can be defined by H in Eq. 8 and create a hierarchy of pcsets in a given harmonic context. The underlying assumption is that pcsets that comply with the user-defined constraints will be predominantly adopted in a composition, thus evoking higher stability due to their larger frequency of occurrence (Bharucha, 1984b). Naturally, harmony is one element within the many perceptual attributes of the musical surface elaboration that convey different stability conditions. The interaction of harmonic content with remaining structural elements – such as rhythmic (e.g. duration), metrical strength, and pitch height–is fundamental to the perception of the stability and the hierarchical formation of musical events.

Table 2 shows the 14 highest ranked pcsets with cardinality $t = 3$ sequentially listed by their decreasing value of harmonic stability H for three user-input parameter conditions. Left and middle data are in $\mathbb{Z}_{12}$, where we explore Western music archetypes for tonal diatonic structures. It adopts as a reference pcset R the diatonic pitch collection of C major, $R = \{3, 0, 1, 0, 2, 1, 0, 2, 0, 1, 0, 1\}$ with the C major triad notes (i.e. C, E, and G) reinforced in terms of *importance* or *stability*. Moreover, the IC distribution $I = \{0, 0, 1, 1, 1, 0\}$ derives from a major/minor triad pcset, thus equally enforcing the complementary intervals of m3/M6, M3/m6, and P4/P5, fundamental to the tonal lexicon. The difference between the first two columns relies on the contribution of each constraint by adopting $\alpha = .3$ (left) and $\alpha = .7$ (middle). It exposes the importance of the α parameter in the resulting harmonic lexicon. Larger values of α privilege the user-defined IC in I, and smaller values of α enforce the pitch class content of the reference pcset R. In Table 2, the left data privileges the retention of the reference pcset R; thus, the upper ranking of pcsets is predominantly in the diatonic set of C major. Conversely, the middle data enforce the ICs I, thus exposing an ambiguous major/minor modes tonal centre yet denoting a strong emphasis on triadic harmony within the region of C.

The rightmost data in Table 2 is in the $\mathbb{Z}_7$ space. It adopts as input parameters $R = \{3, 0, 1, 2, 1, 2\}$, $I = \{0, 1, 1\}$, and $\alpha = .3$.[10] The definition of a reference pcset R and IC distribution I for $\mathbb{Z}_7$ follows some perceptual experiments done by the authors and has been arbitrarily chosen, without relying on any existing set. Our rationale aims to promote different levels of importance to

[9] Event hierarchies results from multiple structural conditions, which include, to cite a few, schematic interpretations against the major/minor tonal hierarchies and by the events' metrical position and frequency of occurrence (Bharucha, 1984a,b).

[10] Please note that the input parameters R and I are independent. They can be derived from any source, such as a pcset or perceptual ratings, and define the relative *importance* of pitch classes in R and ICs in I, thus can assume any (integer or float) value.

Table 2. Harmonic hierarchy for three collections of triads (i.e. cardinality $t = 3$) in the $\mathbb{Z}_{12}$ (left and middle data) and $\mathbb{Z}_7$ (right data), whose ranking results from the harmonic stability indicator H. The $\mathbb{Z}_{12}$ lexica adopt different values of α in Eq. 8, enforcing either the IC ($\alpha = .4$) and the pitch class retention ($\alpha = .7$). The lexica in $\mathbb{Z}_{12}$ adopt $R = \{3, 0, 1, 0, 2, 1, 0, 2, 0, 1, 0, 1\}$ and $I = \{0, 0, 1, 1, 1, 0\}$. The $\mathbb{Z}_7$ lexicon adopts $R = \{3, 0, 1, 2, 1, 2\}$ distribution and $I = \{0, 1, 1\}$. Please note that similar pcsets in $\mathbb{Z}_{12}$ and $\mathbb{Z}_7$ do not adopt the same tuning.

pcset in $\mathbb{Z}_{12}\alpha = .3$	Harmonic stability H	pcset in $\mathbb{Z}_{12}\alpha = .7$	Harmonic stability H	pcset in $\mathbb{Z}_7\alpha = .3$	Harmonic stability H
0, 4, 7	.680	0, 4, 7	.872	0, 3, 6	.618
0, 4, 9	.623	0, 4, 9	.864	0, 3, 4	.526
0, 3, 7	.574	0, 3, 7	.857	0, 2, 3	.513
0, 5, 9	.574	0, 5, 9	.857	0, 2, 6	.513
4, 7, 11	.574	4, 7, 11	.857	0, 4, 6	.513
0, 4, 8	.559	0, 5, 8	.854	0, 2, 4	.462
0, 5, 8	.544	2, 7, 11	.853	0, 3, 5	.462
2, 7, 11	.538	0, 3, 8	.850	2, 3, 6	.461
0, 2, 7	.538	2, 7, 10	.850	3, 4, 6	.446
0, 7, 9	.535	4, 8, 11	.850	0, 1, 3	.446
0, 4, 5	.532	1, 4, 9	.849	0, 1, 6	.423
0, 4, 11	.532	2, 5, 9	.849	0, 5, 6	.423
0, 5, 7	.529	1, 4, 8	.847	0, 2, 5	.404
0, 7, 11	.523	2, 5, 10	.847	1, 3, 6	.404

the set, having the pitch class 0 as a stronger tonal center. Furthermore, we aimed at having some degree of well-formedness, namely symmetry (Carey & Clampitt, 1989). The IC distribution aimed to enforce the more consonant intervals, objectively assessed by Vassilakis' (2001) sensory dissonance model.[11]

Figure 5 shows topological Fourier phase spaces in $\mathbb{Z}_{12}$ (left image) and $\mathbb{Z}_7$ (right image) for the 14 best-ranked pcsets with cardinality $t = 3$. We adopt the same input parameters (I and R) from the harmonic stability ranking in Table 2, and a $\alpha = .3$ in both spaces. To reduce the high-dimensional Fourier phase distances of pcsets to a 2-dimensional topology, we apply multidimensional scaling (MDS). This strategy has been largely conveyed by cognitive psychology research to visualise pitch distances (Miller, 1989). To create an MDS representation from pcsets distances in the Fourier phase space, we first compute a square distance matrix between each pair of pcsets using Eq. 2 solved for θ. Then, we apply a classical MDS algorithm (Torgerson, 1952) to create a 2-dimensional projection, where the between-pcset distances are preserved with minimal distortion (or minimal stress using the MDS terminology).

In the $\mathbb{Z}_{12}$ space, the most striking property is the emergence of fuzzy clusters where functional harmonic categories within the region of C major can be observed. Clusters are identified in the left image of Figure 5 as SD and D labels, standing for subdominant and dominant functional categories, respectively. These clusters emerge primarily from aggregating pcsets that share common pitch classes, an information conveyed by the Fourier phase information. The tonic chord, labeled as T, is identified in Figure 5 and isolated from remaining clusters. Note that some scale degrees of the tonic, dominant, and subdominant chord groups – $(\hat{1}, \hat{3})$, $(\hat{7})$, and $(\hat{4}, \hat{6})$, respectively – are naturally separated in $T(3)$ and $T(4)$ DFT coefficient spaces (Yust, 2019). In part, this is due to the thirds-based structure in these coefficients, associated with hexatonicity and octatonicity harmonic qualities, and it points to a univocal relationship between coefficients $T(3)$ and $T(4)$, and the foundation of functional syntax, often summarised as $ii{-}V{-}I$, i.e. subdominant-dominant-tonic. Furthermore, by crossing the information with the pcset stability H, we can use these topologies to infer possible paths across pcsets in the space, enforcing common pitch classes (loosely linked to voice-leading parsimony).

A similar rationale can be applied to less studied and systematized spaces as the $\mathbb{Z}_7$ shown in the right image of Figure 5. The composer should define clusters and harmonic paths to establish lexical and syntactical relationships within a work. For example, one of the prominent features in $\mathbb{Z}_7$ is the separation between pcsets with 'leading tone' (in this context, the pitch class 6). Conversely to the leftmost pcsets, the rightmost pcsets embeds the pitch class 6. This distribution is similar to the one found in the $\mathbb{Z}_{12}$ universe through the dominant and subdominant groups. In this context, it is of compositional relevance to ask whether successive pendulums between pcsets of the first group and pcsets of the second might constitute prototypical harmonic progressions in $\mathbb{Z}_7$ – as it is usually the case in $\mathbb{Z}_{12}$. Furthermore, the topological representation of the space can quintessentially promote geometric-driven harmonic systems by applying concepts such as rotation, translation, and morphing. A video demonstration of harmonic sequences

[11] The sensory dissonance of the IC1, IC2, and IC3 in $\mathbb{Z}_7$ equal to 5.98, 3.77, and 2.5, respectively.

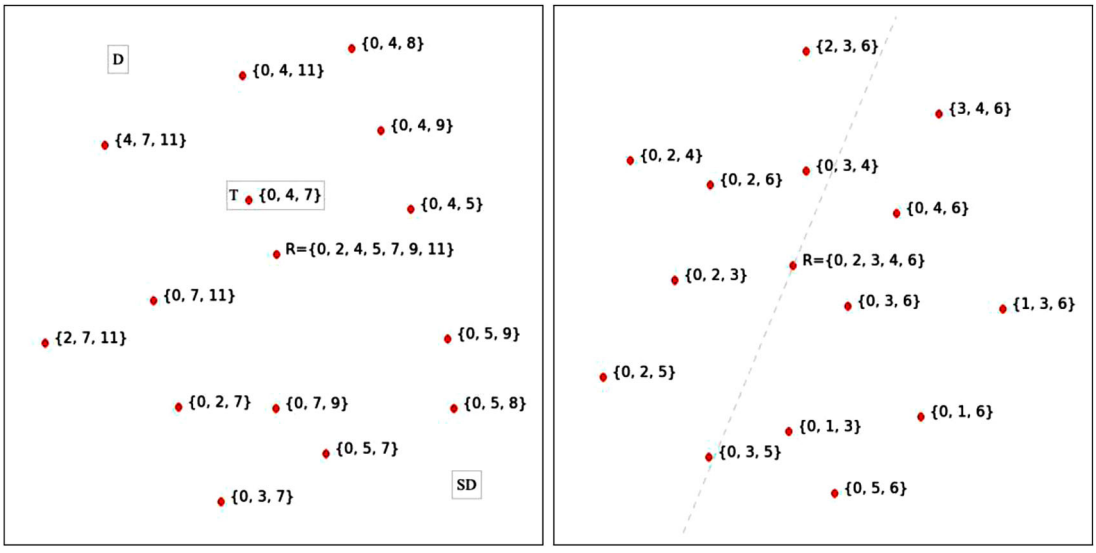


Figure 5. Multidimensional scaling reduction of the Fourier phase space exposing common pitch class relationships for the highest ranked 14 pcsets with cardinality $t = 3$ in $\mathbb{Z}_{12}$ and $\mathbb{Z}_7$ on the left and right images, respectively. The left image adopts adopt $R = \{3, 0, 1, 0, 2, 1, 0, 2, 0, 1, 0, 1\}$ and $I = \{0, 0, 1, 1, 1, 0\}$. The right image adopts $R = \{3, 0, 1, 2, 1, 2\}$ and $I = \{0, 1, 1\}$. Both spaces adopt $\alpha = .3$. In the left $\mathbb{Z}_{12}$ image, we identify the clusters for subdominant (SD), dominant (D), and the tonic chord (T). In the right $\mathbb{Z}_7$ image, a dashed line splits pcsets into two groups based on the embedding of the 'leading tone,' the pitch class 6. Please note that similar pcsets in $\mathbb{Z}_{12}$ and $\mathbb{Z}_7$ do not adopt the same tuning.

defined in the $\mathbb{Z}_{12}$ and $\mathbb{Z}_7$ topological spaces of Figure 5 can be found in the supplementary materials to this article.

Inspired by the hierarchical pitch representation by Lerdahl (1988),[12] Table 3 shows the most stable pitch classes for previously given user-defined parameters from FluidHarmony at different hierarchies (labeled as 'levels'). Each level relates to a cardinality, and higher levels denote increasing cardinalities. It includes pitch configurations based on constructs from 12-TET Western art music, such as the total pitch collection, a regional set (typically including the notes of a scale), chords (triads and tetrads), a dyad, and a single pitch.

Computationally, each pcset per level corresponds to the most stable pcset per cardinality t (the stability of pcsets is given by H in Eq. 8). In other words, the pcsets featured in Table 3 result from computing the pcsets with the highest H from the same set of R and I parameters for the multiple values of $1 \leq t \leq p$. Conversely to existing pitch models (Krumhansl, 1979; Longuet-Higgins, 1987; Shepard, 1982), FluidHarmony can define all levels of pitch description and establish their hierarchical relations. The resulting representation denotes a 'well-formed' basic space as proposed by Lerdahl (2001).[13] Well-formed representations, denoting the basic space of some user-defined conditions, are enforced if smaller values of α are adopted to privilege the regional space, defined by the reference pcset R, over the ICs I.

The basic space shown in Table 3 can be understood as a context-sensitive stratification of a p-TET space. Stability decreases as we move down from level a in the space. Each level is cumulative, as pitch classes presented at the most stable level are repeated at the lesser stable levels. This cumulative property of the levels follows the tradition of the reductional space promoted by the Generative Theory of Tonal Music (Lerdahl & Jackendoff, 1996). While outside of the scope of this article, pursuing a mathematical framework from the proposed representation can expand the music-theoretical and music-psychological discourse of Lerdahl's (2001) tonal pitch space theory beyond the 12-TET Western tonal context. In the context of our proposal, on the use of the FluidHarmony method for assisting in the composition process, the levels can be either explicitly considered as a reference pcset R, or as a guiding framework for guiding the composer in defining micro and macro harmonic objects.

Note that in all of the examples provided in the current section, the reference pcset R typically adopts a large cardinality t within the possible number of pitch

[12] The inspiration for the representation in Table 3 is attributed to Lerdahl's (1988) Tonal Pitch Space, whose roots can be found in Deutsch and Feroe (1981).

[13] The concept of well-formedness has a conflicting definition in the literature. In the context of our work, we follow the definition proposed by Lerdahl (2001). For a comprehensive comparative discussion on the multiple proposals, please refer to Noll (2010).

Table 3. Pitch class hierarchical space representation inspired from Lerdahl (1988), where the embedding of the space across hierarchies is highlighted. Each level is related to a cardinality t, and all possible levels (and cardinalities) are represented. Per level, we notate the pcset with higher stability H computed from the FluidHarmony method with the following parameters. The representation (a) is in Z12 and adopts $R = \{3, 0, 1, 0, 2, 1, 0, 2, 0, 1, 0, 1\}$ and $I = \{0, 0, 1, 1, 1, 0\}$. The representation (b) is in $\mathbb{Z}_7$ and adopts $R = \{3, 0, 1, 2, 1, 2\}$ and $I = \{0, 1, 1\}$. Both spaces adopt $\alpha = .3$. Please note that similar pcsets in $\mathbb{Z}_{12}$and $\mathbb{Z}_7$do not adopt the same tuning.

(a)

card t	level											
1	a	0										
2	b	0						7				
3	c	0				4		7				
4	d	0				4		7		9		
5	e	0		2		4		7		9		
6	f	0		2		4	5	7		9		
7	g	0		2		4	5	7		9		11
8	h	0		2		4	5	7	8	9		11
9	i	0	1	2		4	5	7	8	9		11
10	i	0	1	2		4	5	7	8	9	10	11
11	j	0	1	2		4	5	6	7	8	9	10	11
12	j	0	1	2	3	4	5	6	7	8	9	10	11

(b)

card t	level							
1	a	0						
2	b	0				3		
3	c	0				3		6
4	d	0			2	3		6
5	e	0			2	3	4	6
6	f	0			2	3	4	5	6
7	g	0		1	2	3	4	5	6

classes from a collection p. This strategy follows a traditional *modus operandi* within Western (tonal) art music, where harmonic structures tend to adopt a top-down definition. For example, given a regional context (or a key), we can define subsets that enforce stability further. While this strategy is not a requirement in the FluidHarmony method, it offers greater control over the generated harmonic lexicon. However, the definition of a reference pcset in FluidHarmony can exist at every hierarchy, conveying different degrees of control over pitch class retention from the reference pcset R.

6. Evaluation

This section details an objective computational assessment of our FluidHarmony method in predicting the harmonic lexicon in Western art musical pieces and how it aligns with composition practice across multiple historical periods within the Western art music tradition. Specifically, we are assessing how well the constraints in FluidHarmony model harmonic pcset perform on lexica modelling for individual composers and between groupings of pre- and post-1900 composers. We assume the remaining spaces to be explored, particularly equal temperaments beyond 12-tone subdivisions, can be formally defined from the same constraints.

A representative sample of Western music can be found in the *The Yale–Classical Archives Corpus* (YCAC) (White & Quinn, 2016), whose properties are described in Section 6.1. Given the focus of the current study in promoting CAAC for less systematized contemporary practices, we expanded the YCAC with more representative examples across the twentieth century. To infer the input parameters of the FluidHarmony model, we compute statistics from each corpus' file. Then, we define a pcset ranking given the harmonic stability H value from FluidHarmony. Finally, we quantify the degree to which the proposed FluidHarmony ranking captures a pcsets frequency distribution of each file and bolster intuitive observations on our method from the large amounts of analyzed data.

Furthermore, the implication and contribution of the twofold constraints are studied on the corpus by inferring the value of α in Eq. 8 that best performs in our method. Our expectancy is that both constraints are fundamental to the computation of a harmonic lexicon, namely in the common-practice period. However, it is currently unknown the degree to which each parameter contributes to lexicon formation.

6.1. Corpus properties and annotations

The YCAC yields data from 8,713 MIDI files of pieces or movements from Western European Classical art music. The corpus was compiled from digital musical scores indexed in the crowd-sourced archive classicalarchives.com. Therefore, no systematic criteria exist for the resulting collection, reflecting the priorities of a group of individuals committed to converting their favourite pieces into a digital format. Despite its undeniable value for a systematic analysis of symbolic music, misrepresentations of the musical score have been identified in the YCAC. As DeClerq (2016) notes, the misrepresentations derive mostly from the automatic detection of scale degrees and local keys. However, they are usually 'coherent' with the underlying macroharmony. As our evaluation tackles relations between local and large structures inferred from the first, these misrepresentations may have little influence on our results. White and Quinn (2016) highlight the higher representativeness of 'the usual suspects'–e.g., Bach, Beethoven, Mozart–in the corpus along with some composers from whom little to no information is available. To avoid any poor statistical results that could bias the observation per composer towards particular musical examples, we excluded all composers in the YCAC with less than eight files from

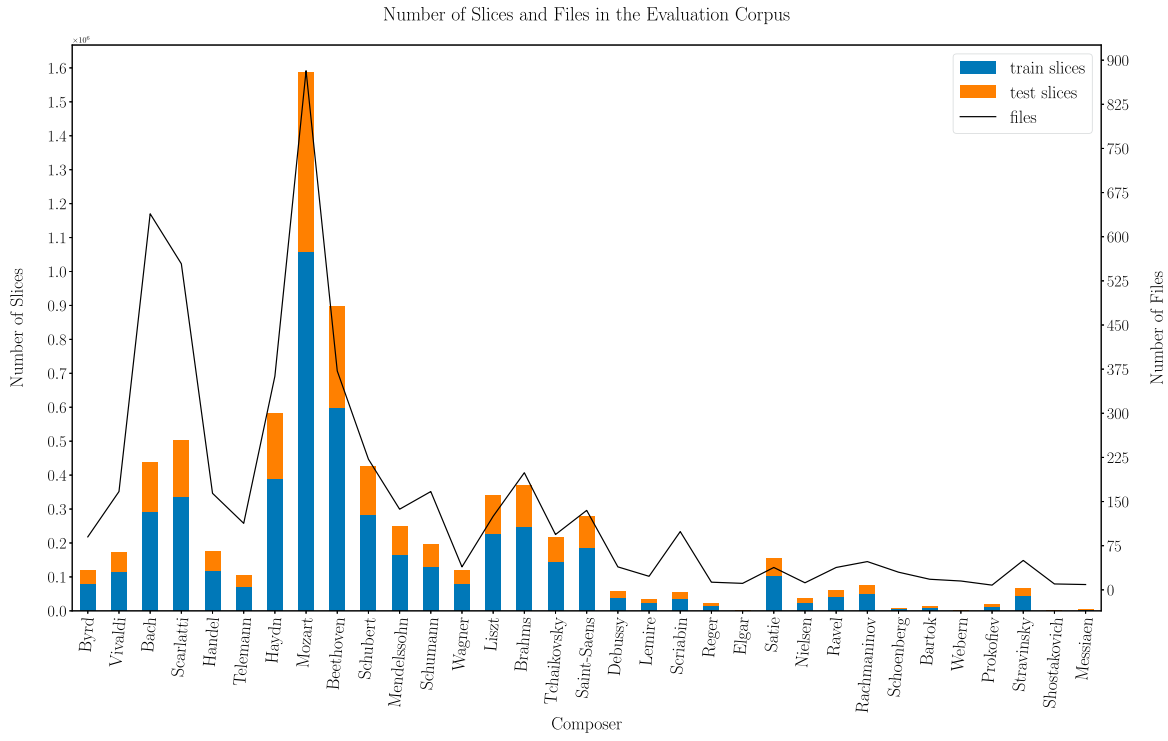


Figure 6. Number of files and slices per composer in the evaluation corpus. Slices are split into train slices (used to parameterise the FluidHarmony method) and the slices adopted for testing the method. Composers are chronologically ordered in the *x*-axis, providing an overview of the distribution of files and slices across the temporal span of the corpus.

our analysis. Moreover, we equally excluded monophonic MIDI files due to the inability to compute an input IC distribution–a fundamental parameter to our method. In total, 5,112 files from the YCAC were processed.

In the resulting corpus, the lack of twentieth century composers was particularly noticeable. Due to the focus on the application of the FluidHarmony method in less systematized harmonic systems outside the Western common-practice period, we expanded the corpus with 72 new files from the following four representative twentieth century composers: Olivier Messiaen, Arnold Schoenberg, Igor Stravinsky, and Anton Webern. A total of 5,184 files were compiled in the corpus, whose distribution per composer is shown in Figure 6. The extended set of files was processed and annotated following the YCAC syntax and can be found in the supplementary material to this article.

Data and metadata annotations for each file in the extended YCAC cover a variety of fields such as pitch and rhythmic content information of the musical surface, title (name of the piece), composer (last name), date of publication, and genre indicating the piece's compositional or formal type (e.g., symphony, character piece, opera, or mass.) For a comprehensive description of available annotations, please refer to White and Quinn (2016).

Fundamental to our evaluation is the corpus data representing the pitch class information per 'salami slice,' i.e. a segmented pcset each time a pitch is added or subtracted from the musical surface.[14] The corpus includes a total of 7,582,130 slices, whose distribution per composer is shown in Figure 6. Given the need to parameterise the DFT space by the input vectors R and I, we created a threefold split of each file in the corpus and adopt 2/3 of the total number of salami slices to compute the input parameters and 1/3 to test the method–a typical split within machine learning evaluation when adopting the same dataset for training and testing (James et al., 2013). The test slices are retrieved from the end of each file, where modulations are less prone to occur, thus minimising possible mismatches between the parameterisation of R and the testing pcsets. From the total number of slices, 5,054,753 are used for parameterising the DFT space (i.e. defining R and I), and the remaining 2,527,376 for testing the FluidHarmony method. Figure 6 shows the split between the number of slices used to compute the parameters and the number of slices used for testing the method per composer.

[14] The salami slice segmentation can be computed from MIDI files by the chordify() function within the music21 software package.

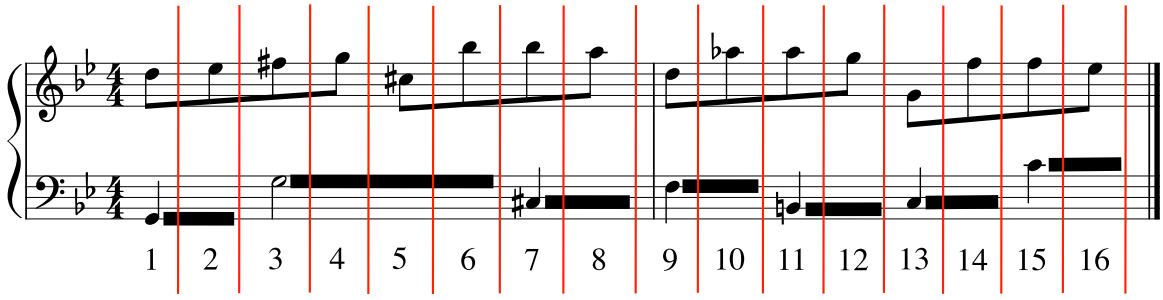


Figure 7. Salami slice segmentation of the first two measures of Bach's Aria *Meine Seufzer meine Tranen* from the *Achzen und erbarmlich Weinen* in G minor, BWV 13. Slices are numbered and denoted by vertical bounding lines.

The salami slice segmentation method offers access to the musical surface without biases towards particular constructs or assumptions, namely on what constitutes a chord or what structures should be privileged. However, this equally means that, for example, pcsets resulting from processes of embellishment and elaboration, such as passing notes, neighbour notes, suspensions, and appoggiaturas, will create segments that do not comply with the underlying harmonic lexicon. Figure 7 exposes how the method can introduce inconsistencies in the resulting evaluation. For example, the slice three in Figure 7 results in the non-diatonic pcset {6, 7} in the context of G minor, as a result of the *appoggiatura* by step-wise motion in the upper voice. While some research has experimented with ways to introduce equivalencies into the alphabet of chord types (White, 2013), the power and peculiarity of the YCAC salami-sliced data should be noted and remains, to the best of our knowledge, the most reliable segmentation strategy to assess our method without assumptions.

6.2. Computational methods

For each file in the corpus, we parsed its annotation file and retrieved its entire collection of pcsets as salami slices. Per file, two distributions were computed from the first 2/3 of the total number of salami slices: an IC distribution and a 12-element pitch class histogram. The resulting vectors are input to our FluidHarmony method as parameters I and R, respectively. The two remaining user-defined attributes, cardinality t and magnitude/phase balance α were set to $1 \leq t \leq 12 \in \mathbb{Z}_{12}$ and $0 \leq \alpha \leq 1 \in \mathbb{R}$, respectively. We instantiate FluidHarmony with all unique pcset combinations $P_i$within Z12, i.e. all combinations of pitch classes in the range $[0, 11] \in \mathbb{Z}$ with pcsets of 1–12 elements or cardinalities. A total of 4,095 unique pcsets combinations are created.

We ran the FluidHarmony method for each file in the corpus and computed its pcset ranking. A frequency distribution histogram for each file is created using the ranked pcsets order as the distribution indexes (two examples are shown in Figure 9). Finally, we compute a *harmonic ranking index* as the median index of the frequency distribution histogram. The smaller the harmonic ranking index, the better the adopted lexicon in each file fits the harmonic pcset ranking from FluidHarmony.

The harmonic ranking index was computed for all values of $0 \leq \alpha \leq 1$ in Eq. 8 with .05 increment values. For each value of α, we calculate the median of the harmonic ranking index values for all files of a given composer in the corpus. The results (per composer and for the entire corpus) aim to show the relative importance of each constraint in capturing the lexicon of existing Western art music. The data valley indicates the best performing α in providing the best alignment between the pcset ranking generated by the FluidHarmony method and the harmonic lexicon adopted by the composers. Valleys at $\alpha = 0$ and $\alpha = 1$ indicate that only the phase or magnitude, respectively, would be relevant to best capture the harmonic lexicon of the corpus files. A valley within limits indicates that both constraints are relevant to compute the lexicon and their contribution.

6.3. Results

Figure 8 shows the harmonic ranking index averaged per composer using values of $0 \leq \alpha \leq 1$ in Eq. 8. The results show a large agreement across the evaluated composers in the global tendency of the data and the index of their valleys, indicating the best performing α. When combining the entire corpus, shown as a dashed line in Figure 8, the valley is found at $\alpha = .5$. The latter value exposes the best performing α for the FluidHarmony method in capturing harmonic lexica of Western art music and provides an 'agnostic' indicator, or a default parameter value, to guide composers in the initial exploration of FluidHarmony. Furthermore, it validates the importance of both

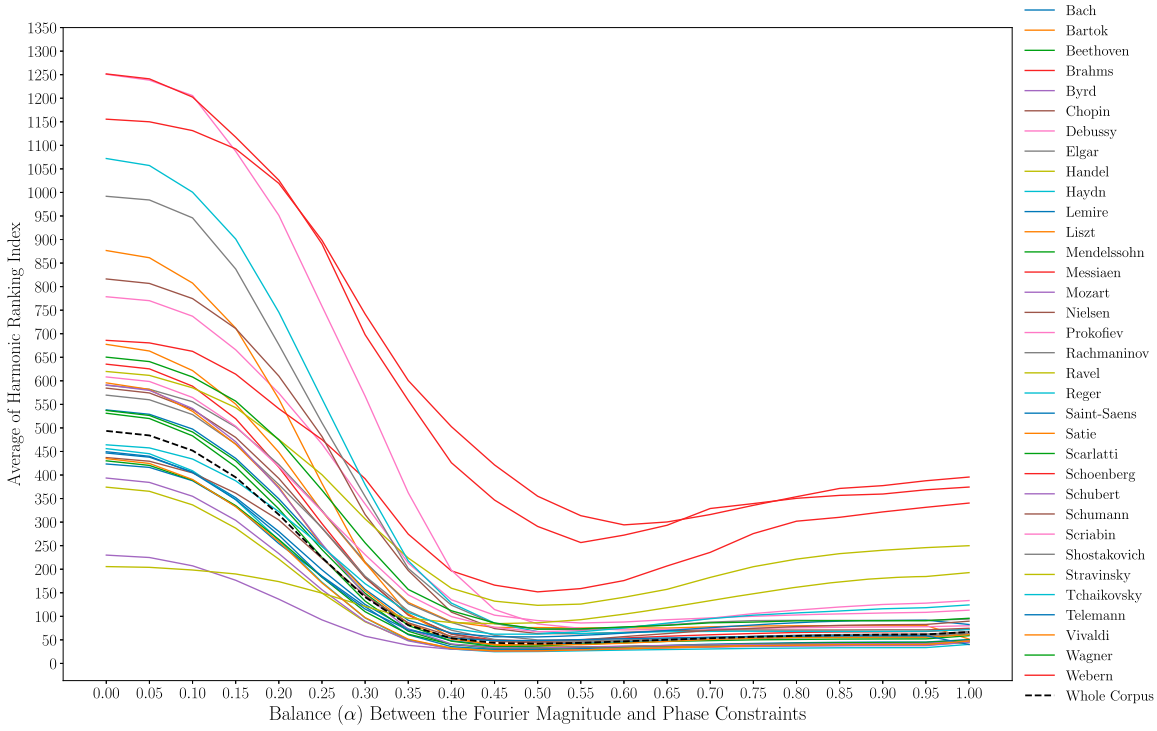


Figure 8. Harmonic ranking median for $0 \leq \alpha \leq 1$ values with .05 step distances in Eq. 8. Results are shown per composer and for the entire corpus (dashed line). The valley $\alpha = .5$ indicates the optimum value to balance both constraints in the proposed method. Balance (α) Between the Fourier Magnitude and Phase Constraints.

constraints in capturing harmonic lexica from the corpus. However, we must note the asymmetry of the results (considering the results for the entire corpus, the skewness is 1.2). It exposes a non-linear relationship between the α range values and the resulting implications in the computed lexicon. By adopting larger α values, thus privileging the magnitude V, we remain closer to the lexicon than privileging the phase C by adopting smaller values of α.

Figure 9 shows the frequency distribution histograms for two files in the corpus by Johann Sebastian Bach. They expose a high (on the left image) and low (on the right image) performing harmonic ranking index for two corpus' files from Bach. The indexes are shown together with the pcset on the y-axis. To enhance the figure's readability, we excluded all indexes from pcsets that were not adopted in a given file. In these two examples, we can observe the implications of the adopted segmentation in the results. In the right image distribution, we can verify the predominance of non-diatonic tones in the lexicon adopted by the composer as the result of verticalities that encompass many ornaments. The entire collection of generated plots are available as supplementary material to this article. A squared box around an index on the y-axis identifies the harmonic ranking index H and provides an intuitive understating of the information captured by this index, namely the relation between the frequency values on the histogram and its placement in the ranked and indexed pcsets.

Descriptive statistics of harmonic ranking indexes for the entire analyzed corpus are shown in Table 4. The descriptive statistics include average and standard deviation (STD) of the corpus' harmonic ranking indexes H. An averaged index of 52.7 ± 110.1 out of the 4,095 unique indexes (or unique pcsets combinations) shows that our Fluidharmony method can capture in the upper pcset ranking the harmonic lexicon used by the composers in the corpus. In other words, the adopted pcset lexicon in the corpus is primarily captured in the 8% top pcsets in the FluidHarmony ranking (we adopt a 86% as a reference for computing the primary pcset information captured by the ranked data in FluidHarmony). Moreover, we shall consider that in most rankings using a $\mathbb{Z}_{12}$ diatonic pitch space, such as those included in the corpus, the initial pcsets in the ranking typically comprise the seven diatonic pitch classes and 21 dyads resulting from the combination of the prominent ICs in I and pitch classes in the reference pcset R. However, within Western art music, a 3- or 4-voice texture is more commonly adopted (Huron, 2001), hence a harmonic ranking index in the [20-50] range is mostly capturing 'in-region' triads and tetrads. Moreover, while the test pcsets are

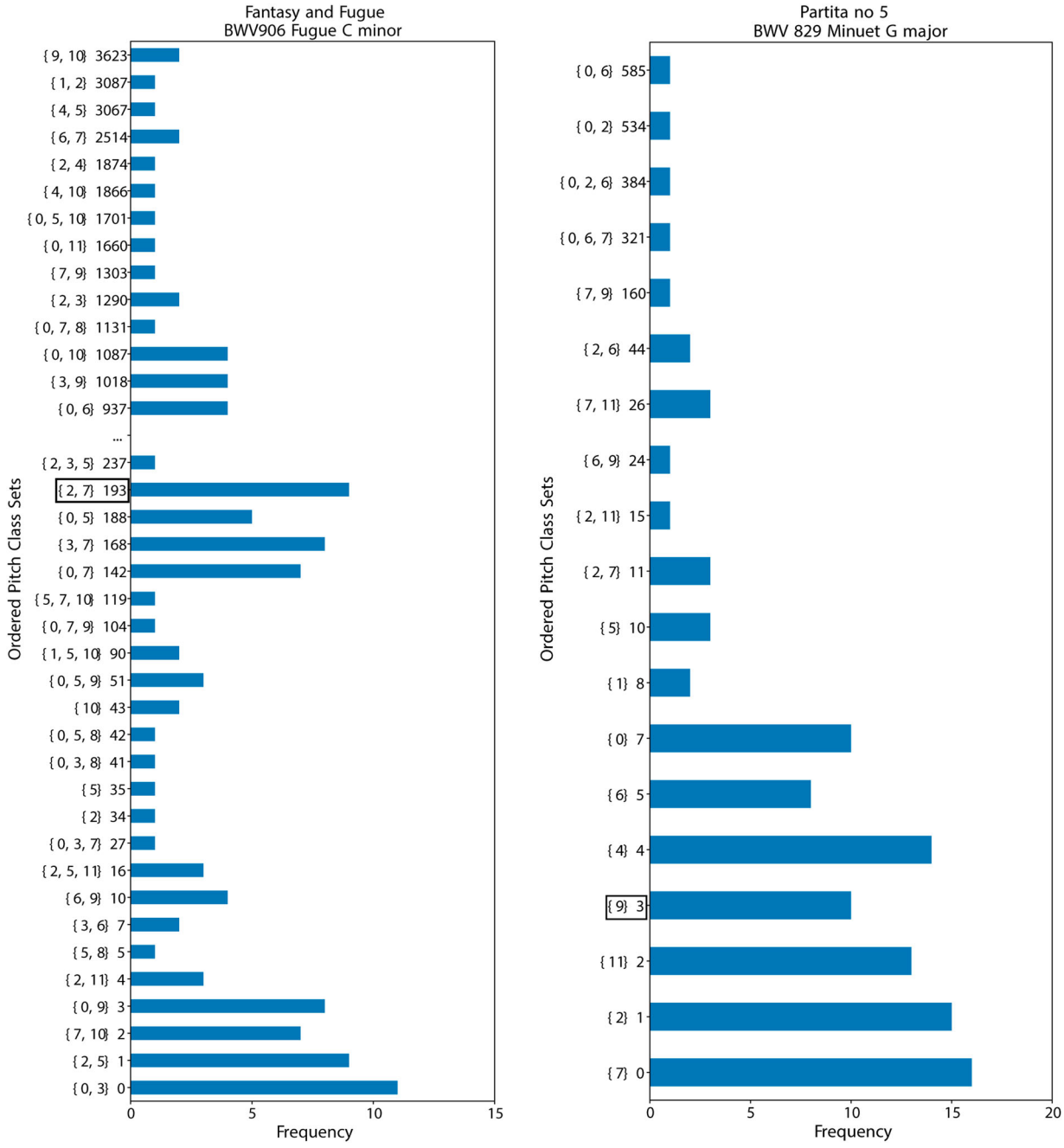


Figure 9. Histogram of pcsets adopted in two compositions by Bach: *Fugue* of the *Fantasy and Fugue* in C minor, BWV 906 (left image) and the Minuet of the Partita no. 5 in G Major, BWV 829 (right image). The order of pcsets in the *y*-axis results from the pcset ranking of FluidHarmony (each label indicates the pcset followed by its ranking index). Index entries without any pcset and, in the right image, multiple entries from the middle of the histogram have been removed for enhanced readability. The distribution's statistics (median and IQR) equals 3 ± 6 and 193 ± 558 for the left and right images, respectively.

retrieved from the last part of each file where modulations are less prone to occur, we must account for some degree of pcsets that are represented to a lesser degree in R, possibly penalising the results with higher harmonic ranking indexes H. While acknowledging this artifact in our evaluation procedure, please note that R accumulates all information from the first 2/3 slices of each piece and enforces the pcsets with greater frequency, which are expected to be in the key of the piece. This possible mismatch can equally have a greater penalty in later historical composers, where modulations or displaced tonal centres are more likely to occur. However, composers after 1900 only account for 10% of the total test slices.

Table 4. Descriptive statistics–average and standard deviation (STD)–for the harmonic ranking index H of the YCAC files under analysis. The analysis presents statistics for the entire corpus and for composers before and after 1900. We observe a gradual increase in the harmonic ranking index in the period after 1900. All analysis adopt a threefold split per file (2/3 for parameterisation and 1/3 for testing).

	Composers before 1900	Composers after 1900	All Composers
Files	4530	654	5184
Slices (param)	4547966	506787	5054753
Slices (test)	2273983	253393	2527376
Average	40.1	140.2	52.7
STD	46.7	268.8	110.1

Table 4 also presents the statistics by grouping composers before and after 1900. This split is defined in the YCAC as 'major' and 'minor' composers, due to their smaller representativeness in the corpus, but equally splits the corpus historically at the turn of the twentieth century. In the context of our work, this split provides a better analysis of the harmonic ranking index H in primarily tonal diatonic lexica before 1900 (40.1 ± 46.7) and more unsystematic approaches to harmony in the twentieth century (140.2 ± 268.8). To better unpack this historically increase in the index H, we show in Figure 10 an analysis per composer. Of note the increasing tendency of the index across historical periods. This aligns with the historical perspective provided by multiple studies conveying the appropriation of a more extensive lexicon and the gradual emancipation of dissonance (Tenney, 1988). Of note are two twentieth century composers a noticeable higher harmonic ranking index H: Schoenberg and Webern. We strongly believe that the worse results by these two composers relate to their adoption of twelve-tone techniques–also known as dodecaphony. The latter approach results in a (nearly) uniform reference pcset R, thus failing to provide information about the region the pcsets should occupy in the Fourier space. In these particular cases, where either the IC distribution I and reference pcset R result in uniform distributions, FluidHarmony constraints may be insufficient in defining a pcset lexicon.

7. Conclusions and future work

We have proposed FluidHarmony, a CAAC method to systematically and intuitively explore the definition of a harmonic lexicon, namely the less systematized equaltempered pitch spaces with any p subdivisions of the octave. In contrast to existing methods (Anders & Miranda, 2010), our method requires little parameterisation by leveraging the Fourier space in eliciting control over the IC and pitch class retention of a reference pcset as high-level concepts, intrinsic to the composition practice within Western art music. The user can control these properties with fine detail by decoupling Fourier magnitude and phase information, respectively. Their contribution to a final ranking of pcsets by harmonic stability can be defined by the user and explored dynamically.

To control the compliance of pcsets to an arbitrarily and user-defined IC distribution, we proposed a strategy to assign weights to the coefficients of a Fourier space algebraically. The strategy is fundamental to our method in controlling the contribution of each IC to the magnitude of a pcset in the Fourier space, used as an indicator of the pcset compliance to the user-defined IC distribution.

The indicator of tonal stability H and the metrics underlying its computation in FluidHarmony, namely the Fourier magnitude and phase information, have been shown to elicit hierarchical representations that can assist composers in defining pcset relations. The harmonic stability indicator provides the most accessible hierarchy that ranks pcsets according to the user-defined constraints. The phase information conveyed by angular distances between pcsets in the Fourier space allows the definition of a novel topology where common pitch classes across sets are expressed as distances, using an MDS algorithm. In mirroring some properties of the $\mathbb{Z}_{12}$ Fourier space, where pcsets within subdominant and dominant harmonic categories exist as fuzzy clusters, we speculate the use of any p-TET phase space to extrapolate the possibility to establish pcset trajectories with syntactic value or foster the exploration of geometric-dependent concepts, such as rotation, translation, and morphing. Finally, a representation inspired by Lerdahl's (2001) tonal pitch space, particularly his basic space with its well-formed property, is conveyed by the FluidHarmony method.

To evaluate our method, we conducted a systematic analysis of a large corpus of 5,064 files (including movements and entire pieces) to assess how well the proposed FluidHarmony method captures the pcset lexicon of existing Western art music. Our corpus included the most represented composers (with eight or more files) in the YCAC (White & Quinn, 2016), which we extended with 72 files by representative twentieth century composers – made available in the supplementary materials to this article. The results provide empirical evidence that FluidHarmony pcset ranking aligns well with the pcset distribution used by the composers in the corpus' files. Our method can predict most of the corpus pcsets in the top 1.3% of the FluidHarmony ranking.

While the FluidHamony method promotes the systematic definition of a harmonic lexicon from IC and pitch class constraints, it does not assure *optimal* surface structure. The embodiment of the resulting lexicon in

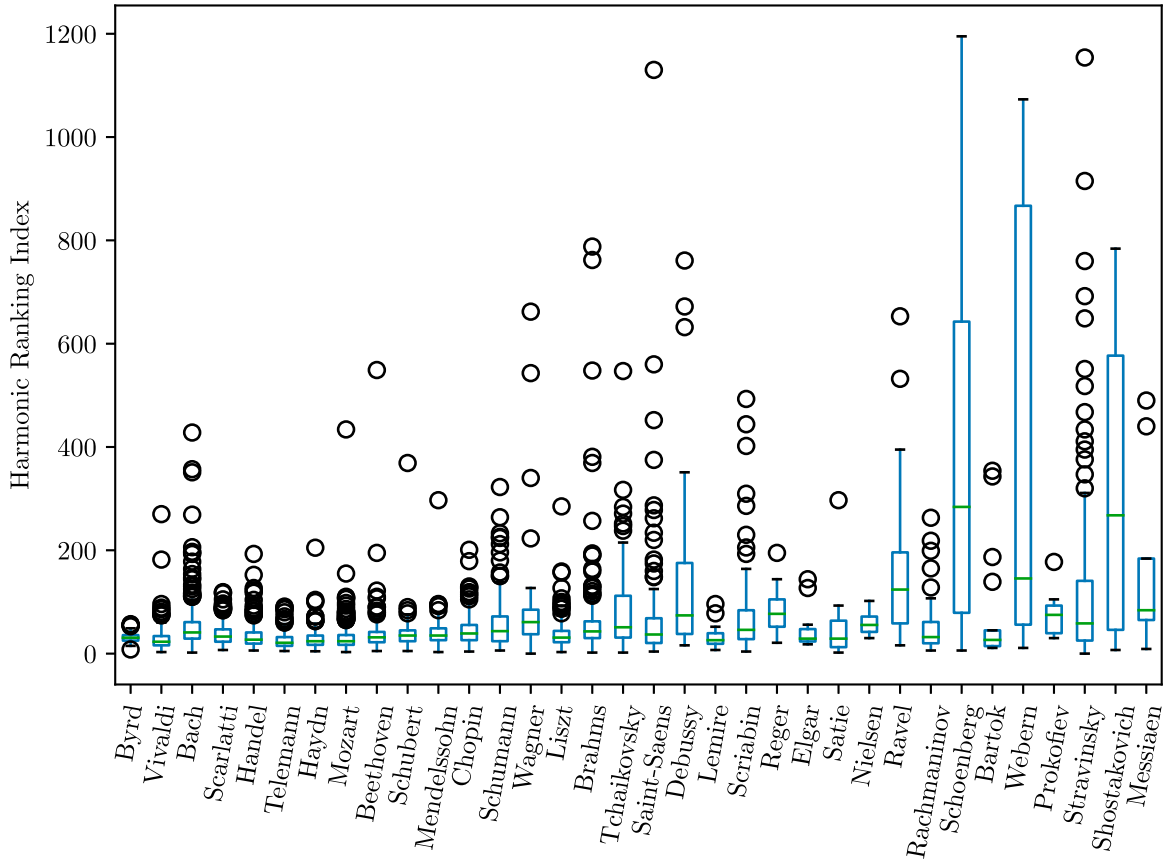


Figure 10. Descriptive statistics (median, IQR, whiskers, and outlier values) for the harmonic ranking index per composer in the corpus for $\alpha = .5$. Composers in the *x*-axis are ordered chronologically from left to right.

musical practice requires a leap from theoretical musical abstractions to surface elaboration, such as the definition of pitch height, rhythmic, and timbre attributes. The surface elaboration can heavily impact the perceptual result and the construction of harmonic coherence akin to event and tonal hierarchies. In the spirit of CAAC tools, FluidHarmony fosters a methodology that precedes the composition practice by offering one approach to the systematic rationalisation of *p*-TET pitch spaces on which prescriptions can be based.

In the supplementary material to this article, we can find manifestations of its many contributions. To promote the creative appropriation and the scientific and artistic extension and reproducibility of our study, we supply two software implementations, the twentieth century musical pieces included in the YCAC, our evaluation data, and a video demonstration of the method implementation in defining pcset sequences.

In the supplementary material to this article, we provide materials that demonstrate the creative applications and reproducibility of our study. Specifically, we provide two software implementations, the twentieth-century musical pieces included in the YCAC, our evaluation data, and a video demonstration of the method implementation in defining pcset sequences. The FluidHarmony method has been implemented in the Python language and Pure Data software environments to reach a broad audience acquainted with either code or visual programming environments.

In the future, the most pressing issue to be pursued is conducting case studies and evaluations protocols that assess the creativity support of FluidHarmony in composition, as well as study the implications of control versus autonomy within the creative practice and the possible appropriations of the method by practitioners. While the FluidHarmony method can be accused of 'universalising' or reducing creative freedom, we are actually aiming to allow greater experimentation by regulated constraints.

The further development of our work would benefit greatly from the wider availability of non 12-TET symbolic music datasets. However, the scarce number of non 12-TET pieces available in the digital domain as symbolic representations in addition to the copyright impediments to make them publicly available, limited our ability to evaluate our method beyond 12-TET. Such an endeavour shall be pursued in the longterm future, as

it can promote the systematic and empirical analysis of twentieth-century Western art music.

We aim to expand further the FluidHarmony method towards a computational cognitive-inspired model, where controlled experimental predictions about mental representations of musical structure, dependency relations, harmonic attraction, and tension in (unfamiliar) *p*-TET pitch spaces, other than 12-TET, can be tested without the same reliance on symbolic datasets. Of interest are the emerging qualities of the proposed hierarchies in eliciting syntactic relations by circular motions orbiting around the reference pcset. The hypothesis related to harmonic motion in the hierarchies proposed, namely those related to the topological representations, can lead to new insights into the mental processes regulating harmonic syntax.

Our findings have a twofold impact at the theoretical and application levels on empirical musicology and music composition. First, we provide evidence that the relative importance of ICs and a reference pcset are fundamental in regulating the harmonic lexicon in 12-TET Western tonal music. The lexicon of Western art music is largely captured by an even combination of these two properties. Second, the proposed weighted DFT space adopting the above elements as spatial constraints and the detailed, quantifiable metrics and hierarchical representations of pitch in any number of equal-tempered octave divisions expand the range of creative support methods and tools available to composers. To our knowledge, such formal methods have never been proposed for pitch spaces beyond 12-TET, and can potentially unlock the definition of harmonic lexicon and leverage novel creative compositional strategies in less studied and systematized pitch spaces.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research has been funded by the Portuguese Foundation for Science and Technology, Research and Technology [2021.05132.BD and 2021.05059.BD] and by the Marie Curie Staff Exchange (MCSE) project "An European and Ibero-American approach for the digital collection, analysis and dissemination of folk music" (Grant Agreement 101086338) under the European Union's Horizon Europe research and innovation programme.

ORCID

Gilberto Bernardes http://orcid.org/0000-0003-3884-2687
Nádia Carvalho http://orcid.org/0000-0001-6882-5138
Samuel Pereira http://orcid.org/0000-0003-2866-1081

References

Aldwell, E., Schachter, C., & Cadwallader, A. C. (2018). *Harmony & voice leading* (5th edition). Cengage.

Amiot, E. (2013). The torii of phases. In J. Yust, J. Wild, & J. A. Burgoyne (Eds.), *Mathematics and computation in music* (Vol. 7937, pp. 1–18). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-39357-0_1.

Amiot, E. (2016). *Music through fourier space: Discrete fourier transform in music theory*. Computational Music Science. Springer International Publishing. Imprint: Springer, Cham, 1st ed. 2016 edition.

Amiot, E. (2017). Interval content vs. Dft. In O. A. Agust´ın-Aquino, E. Lluis-Puebla, & M. Montiel (Eds.), *Mathematics and computation in music* (Vol. 10527, pp. 151–166). Springer International Publishing. https://doi.org/10.1007/978-3-319-71827-9_12.

Anders, T., & Miranda, E. R. (2010). A computational model for rule-based microtonal music theories and composition. *Perspectives of New Music*, *48*(2), 47–77. https://doi.org/10.1353/pnm.2010.0009

Assayag, G., Rueda, C., Laurson, M., Agon, C., & Delerue, O. (1999). Computer-assisted composition at ircam: From patchwork to openmusic. *Computer Music Journal*, *23*(3), 59–72. https://doi.org/10.1162/014892699559896

Bernardes, G., Cocharro, D., Caetano, M., Guedes, C., & Davies, M. E. (2016). A multi-level tonal interval space for modelling pitch relatedness and musical consonance. *Journal of New Music Research*, *45*(4), 281–294. https://doi.org/10.1080/09298215.2016.1182192

Bernardes, G., Davies, M. E. P., & Guedes, C. (2017). Automatic musical key estimation with adaptive mode bias. In *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 316–320, New Orleans, LA. IEEE. https://doi.org/10.1109/ICASSP.2017.7952169.

Bharucha, J. (1984a). Anchoring effects in music: The resolution of dissonance. *Cognitive Psychology*, *16*(4), 485–518. https://doi.org/10.1016/0010-0285(84)90018-5

Bharucha, J. J. (1984b). Event hierarchies, tonal hierarchies, and assimilation: A reply to deutsch and dowling. *Journal of Experimental Psychology: General*, *113*(3), 421–425. https://doi.org/10.1037/0096-3445.113.3.421

Boulanger, R. C. ed. (2000). *The Csound book: Perspectives in software synthesis, sound design, signal processing, and programming*. MIT Press.

Butler, D. (1990). A study of event hierarchies in tonal and post-tonal music. *Psychology of Music*, *18*(1), 4–17. https://doi.org/10.1177/0305735690181002

Callender, C. (2007). Continuous harmonic spaces. *Journal of Music Theory*, *51*(2), 277–332. https://doi.org/10.1215/00222909-2009-004

Carey, N., & Clampitt, D. (1989). Aspects of well-formed scales. *Music Theory Spectrum*, *11*(2), 187–206.

Clough, J., & Douthett, J. (1991). Maximally even sets. *Journal of Music Theory*, *35*(1/2), 93. https://doi.org/10.2307/843811

DeClerq, T. (2016). Big data, big questions: A closer look at the yale–classical archives corpus (C. 2015). *Empirical Musicology Review*, *11*(1), 59. https://doi.org/10.18061/emr.v11i1.5274

Deutsch, D., & Feroe, J. (1981). The internal representation of pitch sequences in tonal music. *Psychological Review*, *88*(6), 503–522. https://doi.org/10.1037/0033-295X.88.6.503

Didkovsky, N., & Burk, P. L. (2001). Java music specification language, an introduction and overview. In *Proceedings of the International Computer Music Conference*, Havana.

Forte, A. (1973). *The structure of atonal music*. Yale University Press.

Ghisi, D., & Agostini, A. (2017). Extending bach: A family of libraries for real-time computerassisted composition in max. *Journal of New Music Research*, *46*(1), 34–53. https://doi.org/10.1080/09298215.2016.1236823

Harte, C., Sandler, M., & Gasser, M. (2006). Detecting harmonic change in musical audio. In *Proceedings of the 1st ACM workshop on Audio and music computing multimedia - AMCMM '06*, 21, Santa Barbara, California, USA. ACM Press. https://doi.org/10.1145/1178723.1178727.

Huron, D. (2001). Tone and voice: A derivation of the rules of voice-leading from perceptual principles. *Music Perception*, *19*(1), 1–64. https://doi.org/10.1525/mp.2001.19.1.1

Huron, D. (2016). *Voice leading: The science behind a musical art* (1st ed.). The MIT Press. https://doi.org/10.7551/mitpress/9780262034852.001.0001.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning: With applications in R*. Springer Texts in Statistics. Springer.

Krumhansl, C. L. (1979). The psychological representation of musical pitch in a tonal context. *Cognitive Psychology*, *11*(3), 346–374. https://doi.org/10.1016/0010-0285(79)90016-1

Laurson, M., Kuuskankare, M., & Norilo, V. (2009). An overview of pwgl, a visual programming environment for music. *Computer Music Journal*, *33*(1), 19–31. https://doi.org/10.1162/comj.2009.33.1.19

Lerdahl, F. (1988). Tonal pitch space. *Music Perception*, *5*(3), 315–349. https://doi.org/10.2307/40285402

Lerdahl, F. (2001). *Tonal pitch space*. Oxford University Press.

Lerdahl, F., & Jackendoff, R. (1996). *A generative theory of tonal music*. MIT Press. repr. edition.

Lewin, D. (1959). Re: intervallic relations between two collections of notes. *Journal of Music Theory*, *3*(2), 298. https://doi.org/10.2307/842856

Lewin, D. (2001). Special cases of the interval function between pitch-class sets x and y. *Journal of Music Theory*, *45*(1), 1–29. https://doi.org/10.2307/3090647

Longuet-Higgins, H. C. (1987). *Mental processes: studies in cognitive science*. Number 1 in Explorations in cognitive science. MIT Press.

McCartney, J. (2002). Rethinking the computer music language: supercollider. *Computer Music Journal*, *26*(4), 61–68. https://doi.org/10.1162/014892602320991383

Messiaen, O. (1944). *Technique de mon langage musical*, Vol. 1. Alphonse Leduc.

Miller, R. (1989). An introduction to multidimensional scaling for the study of musical perception. *Bulletin of the Council for Research in Music Education*, *102*, 60–73.

Milne, A. J., Bulger, D., Herff, S. A., & Sethares, W. A. (2015). Perfect balance: A novel principle for the construction of musical scales and meters. In T. Collins, D. Meredith, & A. Volk (Eds.), *Mathematics and computation in music* (Vol. 9110, pp. 97–108). Springer International Publishing. https://doi.org/10.1007/978-3-319-20603-5_9.

Noll, T. (2010). Two notions of well-formedness in the organization of musical pitch. *Musicae Scientiae*, *14*(1 suppl), 95–113. https://doi.org/10.1177/10298649100140S106

Noll, T. (2019). Insiders' choice: studying pitch class sets through their discrete fourier transformations. In *International Conference on Mathematics and Computation in Music*, pages 371–378. Springer.

Parncutt, R., & hair, g. (2011). Consonance and dissonance in music theory and psychology: Disentangling dissonant dichotomies. *Journal of Interdisciplinary Music Studies*, *5*(2), 119–166.

Perle, G. (1977). *Twelve-tone tonality*. University of California Press.

Persichetti, V. (1961). *Twentieth-century harmony*. W. W. Norton.

Polansky, L., Burk, P., & Rosenboom, D. (1990). Hmsl (Hierarchical music specification language): A theoretical overview. *Perspectives of New Music*, *28*(2), 136. https://doi.org/10.2307/833016

Puckette, M. (1996). Pure data: Another integrated computer music environment. In *Proceedings, Second Intercollege Computer Music Concerts*, pages 37–41, Tachikawa, Japan.

Puckette, M. (2002). Max at seventeen. *Computer Music Journal*, *26*(4), 31–43. https://doi.org/10.1162/014892602320991356

Quinn, I. (2004). *A unified theory of chord quality in equal temperaments*. Phd, University of Rochester.

Quinn, I. (2006). General equal-tempered harmony(Introduction and part i). *Perspectives of New Music*, *44*(2), 114–158. https://doi.org/10.1353/pnm.2006.0010

Quinn, I. (2007). General equal-tempered harmony: Parts 2 and 3. *Perspectives of New Music*, *45*(1), 4–63. https://doi.org/10.1353/pnm.2007.0016

Rohrmeier, M., & Pearce, M. (2018). Musical syntax i: Theoretical perspectives. In R. Bader (Ed.), *Springer handbook of systematic musicology* (pp. 473–486). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-55004-5_25.

Shannon, C. (1949). Communication in the presence of noise. *Proceedings of the IRE*, *37*(1), 10–21. https://doi.org/10.1109/JRPROC.1949.232969

Shepard, R. N. (1982). Geometrical approximations to the structure of musical pitch. *Psychological Review*, *89*(4), 305–333. https://doi.org/10.1037/0033-295X.89.4.305

Simms, B. R. (2000). *The atonal music of arnold schoenberg 1908–1923*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195128260.001.0001.

Susanni, P., & Antokoletz, E. (2012). *Music and twentieth-century tonality: Harmonic progression based on modality and the interval cycles*. Number 1 in Routledge studies in music theory. Routledge, Taylor & Francis Group.

Swain, J. P. (1997). *Musical languages* (1st ed). Norton.

Taube, H. (1997). An introduction to common music. *Computer Music Journal*, *21*(1), 29. https://doi.org/10.2307/3681213

Tenney, J. (1988). *A history of consonance and dissonance*. Excelsior Music Publishing Company.

Torgerson, W. S. (1952). Multidimensional scaling: I. Theory and method. *Psychometrika*, *17*(4), 401–419. https://doi.org/10.1007/BF02288916

Tymoczko, D. (2008). Set-class similarity, voice leading, and the fourier transform. *Journal of Music Theory*, *52*(2), 251–272. https://doi.org/10.1215/00222909-2009-017

Tymoczko, D. (2010). *A geometry of music: Harmony and counterpoint in the extended common practice. Oxford studies in music theory*. Oxford University Press.

Vassilakis, P. (2001). Auditory roughness estimation of complex spectra—roughness degrees and dissonance ratings of harmonic intervals revisited. *The Journal of the Acoustical Society of America*, *110*(5), 2755–2755.

White, C. W. (2013). An alphabet-reduction algorithm for chordal n-grams. In D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi, G. Weikum, J. Yust, J. Wild, & J. A. Burgoyne (Eds.), *Mathematics and computation in music* (Vol. 7937, pp. 201–212). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-39357-0_16.

White, C. W., & Quinn, I. (2016). The yale-classical archives corpus. *Empirical Musicology Review*, *11*(1), 50. https://doi.org/10.18061/emr.v11i1.4958

Yust, J. (2015a). Applications of dft to the theory of twentieth-century harmony. In T. Collins, D. Meredith, & A. Volk (Eds.), *Mathematics and computation in music* (Vol. 9110, pp. 207–218). Springer International Publishing. https://doi.org/10.1007/978-3-319-20603-5_22.

Yust, J. (2015b). Schubert's harmonic language and fourier phase space. *Journal of Music Theory*, *59*(1), 121–181. https://doi.org/10.1215/00222909-2863409

Yust, J. (2016). Special collections. *Journal of Music Theory*, *60*(2), 213–262. https://doi.org/10.1215/00222909-3651886

Yust, J. (2017). Probing questions about keys: Tonal distributions through the dft. In O. A. Agust´ınAquino, E. Lluis-Puebla, & M. Montiel (Eds.), *Mathematics and computation in music* (Vol. 10527, pp. 167–179). Springer International Publishing. https://doi.org/10.1007/978-3-319-71827-9_13.

Yust, J. (2019). Stylistic information in pitch-class distributions. *Journal of New Music Research*, *48*(3), 217–231. https://doi.org/10.1080/09298215.2019.1606833